

~~Solutions~~

1. For each question, credit will be given based on the selected answer(s) only. Do not make stray marks on other answers. If you need to change an answer, fully erase or make sure it is clear what answer(s) you are selecting. No work required.

(1.1) ³ (4 pts) Let $f(x, y) = \sqrt{x+y}$ and $g(x, y) = \frac{1}{x+y}$. Select one answer. The point $(0, 0)$ is a critical point for

- (A) g but not f (B) f but not g (C) both f and g (D) neither f nor g

$(0, 0) \in \text{Domain of } f$

$$f_x = \frac{1}{2\sqrt{x+y}}, \quad f_y = \frac{1}{2\sqrt{x+y}};$$

$$f_x(0, 0) \text{ DNE \& } f_y(0, 0) \text{ DNE}$$

$(0, 0) \notin \text{Domain of } g$

(1.2) ³ (4 pts) Assume $f_x(a, b) = f_y(a, b) = 0$. If $f_{xx}(a, b) = 9$, $f_{yy}(a, b) = 3$, and $f_{xy}(a, b) = -5$. Apply the second derivative test to classify f at (a, b) . Select one answer.

- (A) f has a local maximum at (a, b) . (B) f has an absolute maximum at (a, b) .
 (C) f has a local minimum at (a, b) . (D) f has an absolute minimum at (a, b) .
 (E) f has a saddle point at (a, b) . (F) Second derivative test is inconclusive at (a, b) .

$$D(a, b) = \begin{vmatrix} 9 & -5 \\ -5 & 3 \end{vmatrix} = 27 - 25 = 2 > 0 \quad \& \quad f_{xx} = 9 > 0 \quad \vee$$

(1.3) (6 pts) Consider the conservative vector field $\mathbf{F} = \langle 6xy, 3x^2 + 8y^3z^2, 4y^4z + 3z^2 \rangle$. Identify a potential function for $\mathbf{F}$ if it exists. Select one answer.

- (A) $f(x, y, z) = 3x^2y + 2y^4z^2 + 2$ (B) $f(x, y, z) = \langle 3x^2y, 2y^4z^2, z^3 \rangle$
 (C) $f(x, y, z) = \langle 3x^2y, 3x^2y + 2y^4z^2, 2y^4z^2 + z^3 \rangle$ (D) $f(x, y, z) = 3x^2y + 2y^4z^2 + z^3 + 4$
 (E) No potential function exists. (F) $f(x, y, z) = 3x^2y + 5$
 (G) $f(x, y, z) = 3x^2y + 2y^4z^2 + z^3 + 4z$ (H) $f(x, y, z) = 6y + 24y^4z^2 + 4y^4 + 6z$

$$f(x, y, z) = \int 6xy \, dx = 3x^2y + g(y, z)$$

$$f(x, y, z) = \int (3x^2 + 8y^3z^2) \, dy = 3x^2y + 2y^4z^2 + g_2(x, z)$$

$$f(x, y, z) = \int (4y^4z + 3z^2) \, dz = 2y^4z^2 + z^3 + g_3(x, y)$$

$$f(x, y, z) = 3x^2y + 2y^4z^2 + z^3 + C$$

+ C can include any constant

- (1.4) (6 pts) Consider the integral $\int_0^4 \int_{\frac{1}{2}x}^2 6xe^{y^3} dy dx$. Which of the following is the integral obtained by reversing the order of integration? **Select one answer.**

(A) $\int_0^4 \int_0^{\frac{1}{2}y} 6xe^{y^3} dx dy$

(B) $\int_0^4 \int_{\frac{1}{2}y}^2 6xe^{y^3} dx dy$

(C) $\int_0^2 \int_{2y}^4 6xe^{y^3} dx dy$

(D) $\int_0^2 \int_{2y}^0 6xe^{y^3} dx dy$

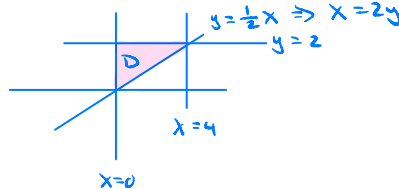
(E) $\int_2^0 \int_{2y}^0 6xe^{y^3} dx dy$

(F) $\int_0^2 \int_0^{2y} 6xe^{y^3} dx dy$

(G) $\int_{\frac{1}{2}x}^2 \int_0^4 6xe^{y^3} dx dy$

(H) $\int_0^2 \int_0^4 6xe^{y^3} dx dy$

D: $0 \leq x \leq 4$
 (vertically simple) $\frac{1}{2}x \leq y \leq 2$
 D: $0 \leq x \leq 2y$
 $0 \leq y \leq 2$



- (1.5) (4 pts) For each of the following vector fields, the domain of $\mathbf{F}$ is $\mathbb{R}^2$ which is simply connected. Which vector fields are **NOT** conservative? **Select all that apply.** No credit will be given if all answers are left blank or all answers are selected.

check if $\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$

☒ (a) $\mathbf{F} = \langle 4x^2 - 4y^2, 8xy - 8y \rangle$ $8y \neq -8y$

☒ (b) $\mathbf{F} = \langle 3x^2 - 4y, 4y^2 - 2x \rangle$ $-2 \neq -4$

☐ (c) $\mathbf{F} = \langle 2xe^{xy} + x^2ye^{xy}, x^3e^{xy} + 2y \rangle$

☒ (d) $\mathbf{F} = \langle 2y^2, (x+2y)^2 \rangle$ $2(x+2y) \neq 4y$

☐ (e) $\mathbf{F} = \langle e^y, xe^y + y \rangle$ $e^y = e^y$

☒ (f) $\mathbf{F} = \langle x^2 - yx, y^2 - xy \rangle$ $-y \neq -x$

☒ (g) $\mathbf{F} = \langle xy, x^2 + x - y \rangle$ $2x+1 \neq x$

☐ (h) $\mathbf{F} = \langle 6 \cos(y), -6x \sin(y) \rangle$ $-6 \sin(y) = -6 \sin(y)$ ✓

(c) $3x^2e^{xy} + x^3ye^{xy} \stackrel{!}{=} 2x^2e^{xy} + x^3ye^{xy} + x^2e^{xy}$ ✓

2. (5 pts) For each true/false question, fill in the appropriate bubble. No work required.

(2.1) (T) ☒ (F) The integral $\int_{-1}^2 \int_0^x x^2y dy dx = \left(\int_{-1}^2 x^2 dx \right) \left(\int_0^x y dy \right)$

(2.2) ☒ (T) ☐ (F) If $\mathcal{D}$ is closed and bounded, then $f(x, y)$ takes on both an absolute maximum and absolute minimum value on $\mathcal{D}$.

(2.3) ☒ (T) ☐ (F) For a differentiable function f , ∇f_P points in the direction of steepest ascent at the point P .

(2.4) (T) ☒ (F) If $\mathbf{F}$ is a vector field, then $\text{div}(\mathbf{F})$ is a vector field.

(2.5) (T) ☒ (F) $\int_0^1 \int_0^z \int_{x+y}^{2x+y} e^{x+y+z} dz dy dx$ is a meaningful triple integral.

3. (8 pts) Fill in the blanks. No work required. No partial credit will be given. All numbers must be simplified and answers must be exact.

(3.1) Assume $\nabla f(P) = \langle 4, -4 \rangle$

The directional derivative of f at P in the direction of $\langle 12, -5 \rangle$ is $\frac{68}{13}$.

$$D_u f(P) = \langle 4, -4 \rangle \cdot \langle 12/13, -5/13 \rangle = \frac{48}{13} + \frac{20}{13}$$

- (3.2) Assuming that $\nabla f(-2, 2) = \langle 48, -12 \rangle$ and $f(-2, 2) = -20$. The equation of the tangent plane to

the graph of f at $(-2, 2)$ is given by $z = -20 + 48(x+2) - 12(y-2)$.

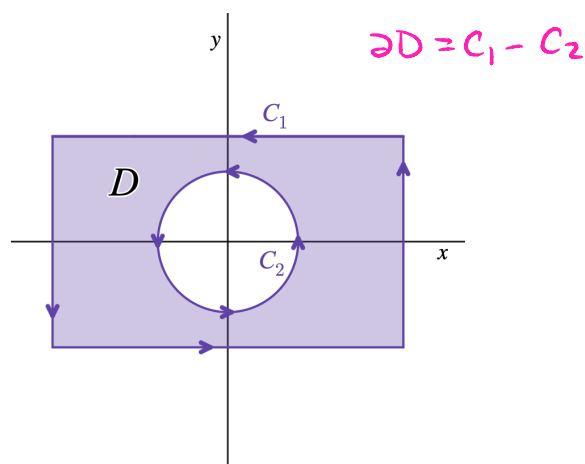
- (3.3) If $\mathbf{F}$ is a conservative vector field and C is a closed curve, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

- (3.4) If $f(x, y) = 3$, then $\iint_D 3 \, dA = 3 \text{ Area}(D)$.

4. (5 pts) Consider the region below. If $\oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = 12$ and $\iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = 80 - 9\pi$, calculate

$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r}$. Simplify your answer and write your final answer on line provided.

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} = 92 - 9\pi$$



$$\begin{aligned} \oint_{\partial D} \vec{F} \cdot d\vec{r} &= \oint_{C_1} \vec{F} \cdot d\vec{r} - \oint_{C_2} \vec{F} \cdot d\vec{r} \\ &= \oint_{C_1} \vec{F} \cdot d\vec{r} - 12 \end{aligned}$$

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = 80 - 9\pi$$

$$\Rightarrow \oint_{C_1} \vec{F} \cdot d\vec{r} - 12 = 80 - 9\pi \Rightarrow \oint_{C_1} \vec{F} \cdot d\vec{r} = 92 - 9\pi$$

5. (6 pts) Let $\mathcal{B} = [-1, 3] \times [1, 2] \times [0, 5]$. Fill in the blanks to compute the triple integral over $\mathcal{B}$ in the order indicated.

$$\iiint_{\mathcal{B}} f(x, y, z) \, dV = \int_{\boxed{A}}^{\boxed{B}} \int_{\boxed{C}}^{\boxed{D}} \int_{\boxed{E}}^{\boxed{F}} f(x, y, z) \, dy \, dz \, dx$$

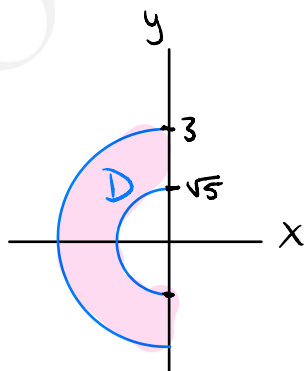
$$\boxed{A} = \underline{-1} \quad \boxed{B} = \underline{3} \quad \boxed{C} = \underline{0} \quad \boxed{D} = \underline{5} \quad \boxed{E} = \underline{1} \quad \boxed{F} = \underline{2}$$

6. (10 pts) Let $\mathcal{D}$ be the region bounded by the circles $x^2 + y^2 = 5$ and $x^2 + y^2 = 9$ in the left half-plane, $x \leq 0$. Fill in the blanks to set up the integral in polar coordinates. **Do not compute the integral.** Hint: Sketching the region of integration might be helpful.

$$\iint_{\mathcal{D}} 6y^3 \, dA = \int_{\boxed{A}}^{\boxed{B}} \int_{\boxed{C}}^{\boxed{D}} \boxed{E} \, dr \, d\theta$$

$$\boxed{A} = \underline{\pi/2} \quad \boxed{B} = \underline{3\pi/2} \quad \boxed{C} = \underline{\sqrt{5}} \quad \boxed{D} = \underline{3}$$

$$\boxed{E} = \underline{6r^4 \sin^3 \theta}$$



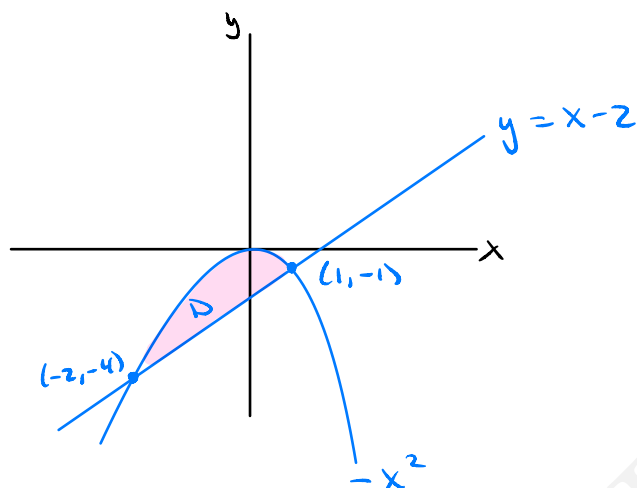
$$\mathcal{D}: \quad \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2} \\ \sqrt{5} \leq r \leq 3$$

$$\iint_{\mathcal{D}} 6y^3 \, dA = \int_{\sqrt{5}}^3 \int_{\pi/2}^{3\pi/2} 6r^3 \sin^3 \theta \, r \, dr \, d\theta$$

7. (12 pts) Evaluate $\iint_D 4x \, dA$, where D is the region bounded by $y = -x^2$ and $y = x - 2$. Show all work and write only your final simplified answer in the space provided. Hint: Sketching the region of integration might be helpful.

Answer: -9

Show work here:



Points of intersection:

$$-x^2 = x - 2$$

$$\Rightarrow x^2 + x - 2 = 0$$

$$\Rightarrow (x+2)(x-1) = 0$$

$$\Rightarrow x = -2, x = 1$$

$$(y = -4) \quad (y = -1)$$

$$D: -2 \leq x \leq 1$$

$$x - 2 \leq y \leq -x^2$$

(vertically simple)

$$\begin{aligned} \iint_D 4x \, dA &= \int_{-2}^1 \int_{x-2}^{-x^2} 4x \, dy \, dx \\ &= \int_{-2}^1 4xy \Big|_{y=x-2}^{y=-x^2} dx \\ &= \int_{-2}^1 (4x(-x^2) - 4x(x-2)) dx \\ &= \int_{-2}^1 (-4x^3 - 4x^2 + 8x) dx \\ &= \left(-x^4 - \frac{4}{3}x^3 + 4x^2 \right) \Big|_{x=-2}^{x=1} \\ &= \left(-1 - \frac{4}{3} + 4 \right) - \left(-16 + \frac{32}{3} + 16 \right) \\ &= -9 \end{aligned}$$

8. (12 pts) Let $\mathcal{C}$ be the curve parameterized by $\mathbf{r}(t) = \langle t^2, e^t, t \rangle$ for $0 \leq t \leq 1$. Complete the set up for the line integrals by filling in the blanks with the **simplified** integrands. Show all work. Do **NOT** compute the integral.

(8.1) $\int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \left\langle 3x, \frac{5z}{y}, -y \right\rangle$

$$\int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \underline{(6t^3 + 5t - e^t)} dt$$

(8.2) $\int_{\mathcal{C}} f(x, y, z) ds$, where $f(x, y, z) = 16z + 4y^2$

$$\int_{\mathcal{C}} f(x, y, z) ds = \int_0^1 \underline{(16t + 4e^{2t}) \sqrt{4t^2 + e^{2t} + 1}} dt$$

Show all work needed to justify your answers here:

$$\int_{\mathcal{C}} f ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| dt$$

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

$$\vec{r}'(t) = \langle 2t, e^t, 1 \rangle$$

$$\|\vec{r}'(t)\| = \sqrt{4t^2 + e^{2t} + 1}$$

$$f(\vec{r}(t)) = 16t + 4e^{2t}$$

$$\vec{F}(\vec{r}(t)) = \left\langle 3t^2, \frac{5t}{e^t}, -e^t \right\rangle$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = \left\langle 3t^2, \frac{5t}{e^t}, -e^t \right\rangle \cdot \langle 2t, e^t, 1 \rangle$$

$$= 6t^3 + \frac{5t}{\cancel{e^t}} \cdot \cancel{e^t} - e^t$$

9. ¹⁰ (8 pts) Let $f(x, y, z) = 4x^2y - 5z$ be a potential function for the conservative vector field $\mathbf{F}$.

(9.1) Calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the path $\mathbf{r}(t) = \langle t, t^3, 4 \rangle$, $0 \leq t \leq 1$. Show all work and write only your final answer in the box provided.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \boxed{4}$$

Show work here:

$$P = \mathbf{r}(0) = (0, 0, 4)$$

$$Q = \mathbf{r}(1) = (1, 1, 4)$$

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= f(Q) - f(P) \\ &= (4(1)^2(1) - 5(4)) - (4(0)^2(0) - 5(4)) \\ &= 4 \end{aligned}$$

- (9.2) ☒ (T) ☐ (F) The value of $\int_C \mathbf{F} \cdot d\mathbf{r}$ will be the same as 9.1 using the path $\mathbf{r}_2(t) = \langle t, t, 4 \rangle$, $0 \leq t \leq 1$.

Justify your answer:

$$P = \vec{r}_2(0) = (0, 0, 4) \text{ and } Q = \vec{r}_2(1) = (1, 1, 4)$$

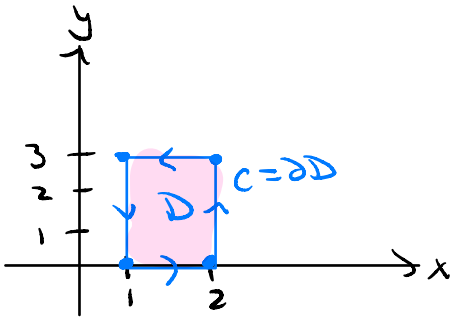
Since $\vec{F}$ is conservative, it is path independent so $\int_C \vec{F} \cdot d\vec{r}$ will be the same for any path from P to Q

10. (10 pts) Use **Green's Theorem** to evaluate the line integral $\oint_C (3x + y^2) dx + (y + 2x^2) dy$, where C is the boundary of the rectangle with vertices $(1, 0)$, $(2, 0)$, $(2, 3)$, and $(1, 3)$ traversed counterclockwise. Show all your work and write only the final simplified answer in the box provided.

Answer:

9

Show work here:



$$\vec{F} = \langle F_1, F_2 \rangle = \langle 3x + y^2, y + 2x^2 \rangle$$

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 4x - 2y$$

$$D: \begin{aligned} 1 &\leq x \leq 2 \\ 0 &\leq y \leq 3 \end{aligned}$$

$$\oint_C (3x + y^2) dx + (y + 2x^2) dy = \int_0^3 \int_1^2 (4x - 2y) dx dy$$

$$= \int_0^3 (2x^2 - 2xy) \Big|_{x=1}^{x=2} dy$$

$$= \int_0^3 ((8 - 4y) - (2 - 2y)) dy$$

$$= \int_0^3 (6 - 2y) dy$$

$$= (6y - y^2) \Big|_{y=0}^{y=3}$$

$$= (18 - 9) - (0 - 0)$$

$$= 9$$